COGNOSCENTI LABS LLC
TECHNICAL NOTE

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Parasitic Communications in Fully Contested Electromagnetic Environments
On the use of hostile emitters as reference fields for covert LPI/LPD links

"Any sufficiently advanced technology is indistinguishable from magic."
Arthur C. Clarke, *Profiles of the Future* (1962)

This note addresses a specific objection: that using a jammer signal as part of a covert communication system is physically unrealistic because jammers destroy signal-to-noise ratio rather than create usable carriers. That objection is correct only if one assumes a conventional receiver trying to decode through interference in the ordinary way. It is not correct for the class of systems discussed here.

The central claim is narrower and more technical. A hostile jammer need not be treated solely as an obstacle. Under bounded conditions, it can be treated as a structured third-party emitter whose spatial, spectral, and temporal characteristics are estimable over useful intervals. In that regime, the jammer's field can provide cover for a low-power LPI/LPD link, and in some cases a usable reference process for estimation, cancellation, and covert signaling. What follows explains that claim using both the patent literature underlying this work and the open academic literature on covert communication and anti-jamming systems.

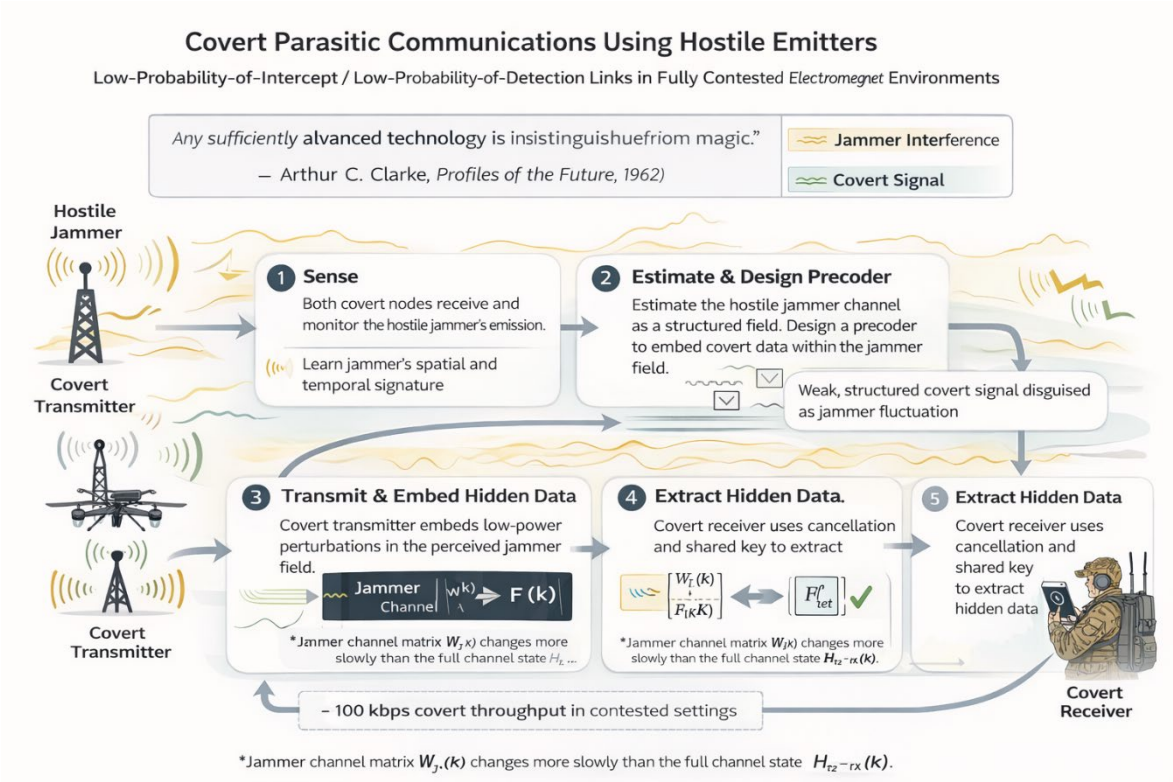
1. The Objection, Precisely Stated

The objection can be stated simply: jammers destroy SNR; they do not create carriers. What exists in practice, one might argue, is spoofing, protocol exploitation, or waveform-specific attack, all of which depend on access to the target waveform or to implementation vulnerabilities.

That objection assumes the receiver is attempting to demodulate the jammer's emission as if the jammer itself were the information-bearing signal. That is not the mechanism at issue here. In a parasitic communication architecture, the jammer is not decoded as content. Rather, it is treated as a dominant field within which the covert system hides low-power, pre-coded perturbations that are only recoverable by a receiver with shared channel knowledge, appropriate cancellation methods, and the correct decoding structure.

The distinction matters. The jammer's power is not the message. It is the cover. The relevant question is therefore not whether a conventional receiver can communicate through a jammer, but whether a purpose-built receiver can estimate enough structure in the hostile field to suppress part of it locally and recover a deliberately embedded low-power signal that remains indistinguishable from ordinary variation to all other observers.

2. Parasitic Communication as the Baseline Concept



Parasitic communication, in the sense developed in the patent family underlying this work, treats an ambient structured emitter as a host signal rather than introducing a stand-alone waveform that is easy to isolate. In OFDM-like settings, the covert transmitter reuses the existing structure of the host environment by introducing carefully shaped perturbations at the symbol, subcarrier, or spatial level. These perturbations may take the form of small phase or amplitude adjustments, per-subcarrier weightings, or spatial precoding operations designed to remain statistically similar to ordinary channel variation and implementation impairment when viewed by an observer without the relevant side information.

Within this framework, the physical channel between the covert nodes serves more than one role. It acts as a propagation path, but also as an authentication surface and, where reciprocity can be exploited, as a source of correlated observations for key generation. A receiver at a different spatial position does not observe the same channel realization and therefore cannot reconstruct the same keying material or identify the same synthesized perturbations. This is why the concept differs from spoofing and from straightforward protocol exploitation. It does not depend on access to a friendly waveform design, nor does it require the presence of a protocol defect. It depends on the ability to characterize an ambient structured field well enough to hide beneath it and decode only at the intended endpoint.

3. Why a Hostile Jammer Can Matter

The step from ordinary parasitic communication to hostile-emitter operation is not a change in physics. It is a change in the source of the ambient field. In cooperative or neutral settings, the host may be a broadcast signal, a cellular downlink, or another structured network emission. In contested settings, the dominant structured emitter may instead be a jammer.

This is the point on which the objection usually fails. A jammer is often described as "noise," but many practical jammers are not noise in the strict sense relevant here. Barrage jammers often have stable spectral occupancy over short intervals. Reactive jammers have detectable response characteristics. Smart jammers may vary over time, but they still exhibit measurable spatial and temporal structure that a sufficiently capable receiver can estimate, at least locally and over bounded windows. The issue is not whether the jammer is friendly, but whether it is characterizable enough for estimation and suppression methods to operate.

There is also an adversarial tradeoff worth noting. A jammer that becomes extremely unstable, highly discontinuous, or difficult to estimate over short windows may indeed complicate covert reception. That same instability can also reduce jamming effectiveness against some receivers and waveforms. In practice, jamming systems occupy a trade space, not a single ideal point. That trade space creates opportunities for systems designed to exploit structured interference rather than simply endure it.

4. Relationship to the Open Literature

The open academic literature supports the broader proposition that a separate emitter's interference can provide cover for covert communication. Sobers, Bash, Guha, Towsley, and Goeckel analyze covert communication in the presence of an uninformed jammer, showing that the jammer's fluctuations can degrade a warden's detection capability while still permitting communication to the intended receiver under appropriate assumptions. The key significance of that work is not the exact numerical result, but the operational model: a third-party emitter creates interference that helps conceal the covert transmission rather than merely obstruct it.

Related work extends this framing to uninformed, informed, and cooperative jammer cases, showing that when the jammer's behavior or channel input is sufficiently known, the legitimate receiver can design cancellation or successive interference strategies around it. Anti-jamming MIMO literature makes the point from a more engineering-oriented direction. There, the jammer is treated as a structured spatial process whose directionality, covariance, or channel ratios can be estimated and used to design beamformers or projection filters that reduce jammer energy while preserving the desired link.

These literatures do not prove every form of hostile-emitter parasitic communication. They do, however, establish that the governing idea is not absurd: a strong interfering emitter can function as cover, and under the right assumptions its structure can be estimated well enough to support receiver-side suppression and purposeful signaling beneath the interference floor.

5. System Interpretation

The architecture relevant here can be understood in four layers.

At the RF front end, both covert nodes receive the hostile jammer's emission. The front end must remain linear enough to operate in the jammer's presence, so analog protection and suppression measures are needed before digitization.

At the analog stage, cancellation and notch behavior reduce front-end overload and protect dynamic range. The purpose here is not perfect removal of the jammer but controlled reduction to a level that makes downstream estimation and processing feasible.

At the digital stage, jammer covariance, angle-of-arrival information, and related statistics are used to construct a spatial or spatiotemporal canceller. This yields a partially cleaned signal in which the jammer residual is lower, but more importantly, more structured and more manageable.

At the baseband layer, the receiver uses channel estimates to interpret a low-power perturbation imposed by the covert transmitter. That perturbation is designed to remain well below the jammer floor for observers lacking the relevant cancellation and decoding structure. The intended receiver, by contrast, sees not the full jammer power but the residual after suppression, and therefore faces a conventional though degraded detection problem.

A compact signal model illustrates the point. At a given subcarrier or narrowband slice, the received signal may be written as the sum of three terms: jammer contribution, covert transmission contribution, and noise. The covert transmitter's energy is deliberately kept small compared with the jammer contribution. After receiver-side suppression, however, the effective observation is dominated not by the full jammer power but by the residual left after cancellation, together with the desired covert term and background noise. The communication problem therefore changes form. It is no longer "decode in the full presence of the jammer," but "decode after structured suppression using a receiver designed for that field."

6. Throughput and Operating Expectations

This is not a claim about universal high-rate communication under arbitrary jamming. That version would be unserious. The more realistic claim is that low-rate covert communication can be sustained under some contested conditions, with burst opportunities during favorable windows.

Published jammer-assisted covert communication results sometimes cite rates in the 10 to 15 Mbit/s range, but those figures generally assume controlled geometry, simplified channel models, stable interference behavior, and strong channel-state assumptions. They should not be carried over uncritically into real contested environments.

In the operating regime of interest here, which may include multiple hostile emitters, non-Gaussian interference, mobility, intermittent nulls, and front-end stress, sustained throughput on the order of 100 kbit/s is a more defensible planning figure. Short bursts above that level may be possible when geometry and interference align favorably. Momentary outages, reacquisition

events, and buffering requirements should be expected as normal design conditions rather than treated as pathologies. The goal is not to mimic a smooth commercial data link. The goal is to retain covert, resilient command-and-control and limited sensor-data exchange in environments where conventional assumptions fail.

7. Operational Relevance

The practical value of this approach appears most clearly in environments where the usual alternatives are absent or constrained.

A dismounted operator communicating with an ISR drone in a heavily jammed band does not necessarily require a broadband link. Sparse tasking, metadata, brief state updates, and occasional short sensor bursts may be enough. A low-rate covert link that survives in the jammer field can therefore be operationally useful even when it falls far below commercial throughput expectations.

A second case arises in the deployment of low-power distributed sensors. If the sensors can use the ambient hostile field as cover for very low-power coordination and data exfiltration, they become more difficult to detect and target as active emitters. In that sense, the jammer field can function not only as an obstacle but as a distributed concealment medium.

A third case is the full-denial environment in which cooperative infrastructure is absent, and the hostile emitter is the only persistent structured source in the band. In that regime, the jammer becomes part of the medium itself. The design question is no longer how to remove all interference, but how to derive enough structure from it to support bounded covert exchange.

8. On the Ukraine Observation

It is true that recent conflict has driven renewed interest in fiber links, greater autonomy, and non-RF control pathways. That observation is correct and important. It is not, however, a technical argument against the approach described here.

Fiber-optic links solve some denial problems while introducing others, including tether dependence, deployment complexity, and range constraints. Laser links can be extremely effective in some cases, but they impose their own demands in line-of-sight, pointing, stability, and weather sensitivity. These are engineering choices across a trade space. They do not eliminate the relevance of covert RF-layer techniques, especially for dismounted users, distributed sensors, or denial scenarios where other infrastructures are unavailable or impractical.

It is also worth noting that the ideas discussed here substantially predate the current conflict. The present argument is not an attempt to retrofit theory to recent events. It is an effort to explain, in light of current literature, why a design direction explored earlier remains technically coherent.

9. Closing

This note was written in response to a specific technical objection. That objection was useful because it forced the issue into precise form: whether a hostile emitter can be treated not merely as interference, but as a structured field that may be estimated, partially suppressed, and exploited to support covert communication. The answer offered here is yes, under bounded and technically realistic conditions.

The claim is deliberately narrow. It is not that every jammer becomes a robust communications backbone. It is that when a hostile emitter is sufficiently structured over a useful interval, and when the receiver architecture is designed for estimation and suppression in that regime, the same field that degrades conventional receivers can provide cover and partial reference for a low-rate LPI/LPD link.

That proposition is consistent with the patent literature cited here, with the open literature on jammer-assisted covert communication and anti-jamming MIMO systems, and with the broader engineering logic of parasitic communication. In fully contested electromagnetic environments, it is not enough to ask only what interference destroys. One must also ask what structured interference makes possible.

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